

Q.4. What is POS tagging? Why it is important?

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• Part-of-speech (POS) tagging is a core task in Natural Language processing (NLP) that involves assigning a grammatical category (or tag) to each word in a sentence.

- These categories describe the syntactic role of a word in the context of a sentence, such as whether it is noun, verb, adjective, adverb, etc.

* Key concepts in POS tagging

• Parts of speech: These are categories like nouns, verbs, adjectives, adverbs, etc. that define the role of a word in a sentence.

• Tagging: The process of assigning a specific part-of-speech label to each word in a sentence.

• Corpus: A large collection of text data used to train POS taggers.

* Types of POS tagging :-

1) Rule based POS tagging :-

uses handcrafted linguistic rules to assign tags.

2) Stochastic POS tagging :-

uses probabilistic models (eg. HMM) to predict tags.

3) Transformation-based POS tagging :-

- combines rule based & statistical approaches.

a) Neural POS tagging

* Why is POS tagging important?

→ • Syntactic Analysis :-

POS tags help in understanding the grammatical structure of a sentence, which is essential for tasks like parsing.

• Example :-

• Sentence : "The cat sat on the mat"

• POS Tags : The / DT cat / NN

sat / VBD on / IN the / DT mat / NN

• Analysis :- The tags show that "cat" is noun (NN), "sat" is a verb (VBD)

and "on" is preposition (IN) & the is Determiner (DT).

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This helps in building a sparse tree for syntactic analysis.

e) Semantic understanding :- knowing the role of a word in a sentence aids in understanding its meaning.

A Example :-
Sentence : "I saw the man with the telescope being used to see the man."

• Example :-

• sentence :- "I saw the man with the telescope".

• Ambiguity :- Is the man using the telescope, or is the telescope being used to see the man?

• POS Tags : I / PRP saw / VBD the / DT man / NN & with / IN the / DT telescope / NN.

• Analysis : The POS tags help clarify that "with the telescope" is a prepositional phrase modifying "saw", suggesting that the telescope was used to see the man.

3) Disambiguation :- Many words in natural language are ambiguous and can belong to multiple pos